

# Guardian AI - Data Sources & Model Architecture

## 1. Guardian AI - Fraud & Scam Detection Chatbot

Data Sources:

- Public datasets: Scamwatch, PhishTank, Spamhaus, FTC Scam Data
- Internal data: Fraud messages, fake job offers, phishing links
- Media data: Fake documents, screenshots, and invoices
- Each entry labeled (Phishing, Fake Job, Crypto Scam, etc.)

AI Models:

1. Fraud Classification Model - Detects scam type
2. Trust Scoring Model - Computes fraud probability (0-100)
3. Global Scam Check Model - Cross-verifies with scam databases
4. Document & Image Authenticity Model - Detects fakes
5. Deepfake Detection Model - Flags AI-generated media
6. Explanation Model - Generates human-readable reasons

Frameworks & Infra:

FastAPI, Flutter, React, MongoDB, SageMaker, CloudWatch, JWT, SSL

## 2. Guardian AI - Research & Analytics Model

Data Sources:

- Social Media (Twitter, Reddit, YouTube)
- News & Financial (Reuters, FiQA, Bloomberg)
- Guardian AI Internal Logs
- Datasets: GoEmotions, HateXplain, Jigsaw

Core AI Models:

1. Sentiment & Emotion Analysis - BERT, RoBERTa
2. Toxicity Detection - ALBERT, CNN+Transformer
3. Trend Forecasting - LSTM, Prophet

4. Topic Clustering - BERTopic, LDA
5. Behavioral Anomaly Detection - Isolation Forest, GNN
6. Market Intelligence - FinBERT, ARIMA
7. Research Report Generator (NLG) - GPT, T5, BART
8. Cross-Domain Correlation - Bayesian Networks

Stack & Architecture:

Kafka, PyTorch, TensorFlow, FastAPI, Airflow, React, D3.js, AWS ECS

### 3. Unified Guardian AI Ecosystem

Architecture Overview:

Input Layer -> Processing Layer -> Insight Fusion -> NLG Reports -> Dashboards

Unified Components:

- Data Input: Text, images, documents, behavioral logs
- AI Processing: Fraud Detection + Analytics Models
- Insight Layer: Correlation + Report Generation
- Interface Layer: Chatbot + Web Dashboard

Real-Time Impact:

- Fraud prevention and scam advisory
- Predictive insights and market trends
- Dynamic AI-generated research summaries